KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION

KARE CAMPUS HELPDESK BOT

Project ID: PRJ033

Team ID: T033

Team GitHub Repository:
https://github.com/Kishore2220/kishore-team-gen-ai-project-

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Project Trainer / Guide

Mr. SHIRWIN PRINCE I

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1.Introduction

Large Language Models (LLMs) write fluent answers, but they can invent facts ("hallucinations") and know nothing about private or recently changed information such as a college's own fees, timings and rules. Retrieval-Augmented Generation (RAG) reduces this problem. Before the model answers, the system fetches the relevant data from an external knowledge base and gives it to the model as the only source for the answer.

This project applies RAG to the KARE Help Desk, a Flutter mobile app that students of Kalasalingam Academy of Research and Education use to report campus issues and look up campus information. The KARE Help Bot answers questions about fees, course subjects, hostels, canteens, campus blocks, the library, exams, medical help and bus routes. Its knowledge base is the information stored in Cloud Firestore. For every question the app retrieves the best matching pages and courses, adds them to the prompt as context, and a Gemini model writes a short answer from that context only.

Each answer ends with a "Based on" line that names the pages it used. When the stored data has no answer, the bot says "Update soon" instead of guessing. If the AI service is unavailable, the bot shows the stored text it found. Because admins edit the pages inside the app, the bot's answers stay current without releasing a new version of the app.

2.Problem Statement & Objectives

Problem: Campus information is scattered. A new student who wants a course fee, a hostel detail, a canteen, a block location or an exam rule has to ask seniors, visit an office or search through separate notices. A plain list or keyword search needs the exact word, and a general chatbot with no access to KARE data can give confident but wrong answers.

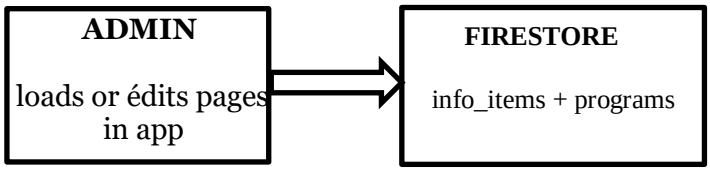
Objectives:

- To build a chatbot inside the KARE Help Desk app that answers campus questions from KARE's own stored information.
- To retrieve the most relevant pages and courses from Cloud Firestore for each question.
- To reduce hallucination by limiting the LLM to the retrieved context, answering "Update soon" when data is missing, and showing which pages the answer is based on.
- To keep the bot useful when the AI service is unavailable, by falling back to the stored text.
- To let admins update the knowledge from the app itself, with no new app release.

3.System Architecture (RAG Pipeline)

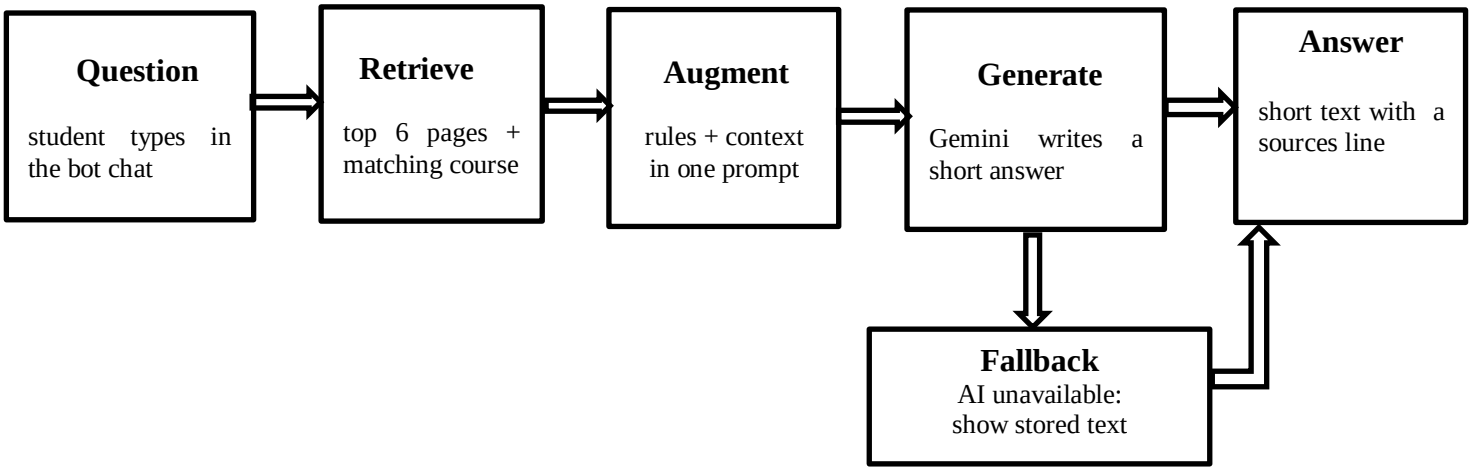
The architecture has two phases: Data Ingestion and Query Processing.

Phase 1: Data ingestion



Each information page and each course is one chunk the bot can retrieve.

Phase 2: Query and answer



3.1 Data Ingestion Phase

The knowledge base is two Firestore collections, not a set of PDFs. info_items holds the hostels (MH1 to MH7 and L1 to L5), canteens, campus blocks, exams, library, medical, transport routes, events and IT pages. programs holds 11 courses with yearly fees and year-wise subjects. Each page and each course is one record, so one record is one chunk and no text splitting is needed. An admin loads the starter data once with the Load starter data button and later changes records with the pencil, Add and Delete buttons. The bot sees every change on the next question.

3.2 Query & Generation Phase

Greetings, thanks and "report an issue" are answered directly. Every other question goes through four steps:

1. **Retrieve:**The question is cut into simple words: lower case, common words removed, plurals trimmed, and synonyms such as "cost" and "tuition" mapped to "fee". Each record is scored (a word in the title scores 3, a word in the details scores 1, a word matching the record's category scores 1) and the top 6 records are kept. A course named in the question, such as CSE or MBA, is matched the same way and its fees and subjects are added.
2. **Augment:** One prompt is built from the rules, the retrieved records, a list of all course fees, a list of all topics in the app and the last few chat messages.
3. **Generate:** The prompt is sent to a Gemini model through Firebase AI Logic. The rules tell the model to take campus facts only from the context, to say "Update soon" when the answer is missing, and never to guess timings, prices, phone numbers or dates.
4. **Respond:** The app shows the answer with a "Based on" line naming the pages used. If the AI call fails, it shows the best stored text from step 1 instead.

4.Implementation Details & Tech Stack

Component	Technology Used	Purpose
Programming language	Dart with Flutter	One code base for the Android app and the Chrome web version
Authentication	Firebase Authentication	Login with register number and password
Knowledge base	Cloud Firestore (`info_items`, `programs`)	Stores campus information and courses; edited by admins in the app
Retrieval	Weighted keyword scoring written in Dart	Finds the top matching records for a question
Large Language Model	Gemini, model `gemini-3.8-flash`, through Firebase AI Logic	Writes the answer from the retrieved context
Security	Firestore security rules	Signed-in users read; only admins edit information

Distribution	Android APK and a private GitHub repository	Install on phones and share the source code

4.1 Retrieval Chunking Configuration

The values below are set in `lib/screens/chatbot_screen.dart`:

```
word in record title            +3
word in record details          +1
word matches record category    +1
records sent as context         top 6
courses sent in detail          up to 3
chat history sent               last 6 messages
AI time limit                   25 seconds
model name                      kBotModel =
                                gemini-3.8-flash
```

4.2 Prompt Rules

The prompt that wraps the retrieved context contains these rules:

- For facts about KARE (fees, courses, subjects, hostels, canteens, blocks, exams, library, medical, transport), use only the context.
- If the context does not hold the answer, or says "Update soon", reply that the information is not added yet and the student should ask the department or office. Never guess timings, prices, phone numbers or dates.
- General study help, such as a project report outline or competitive exam tips, may come from the model's own knowledge and must start with "General advice (not official KARE information):".
- A student who wants to report a problem is told to use Report an Issue on the home screen.
- Answer in simple English, in under 120 words, in plain text.

5.Results and Testing

The retrieval step was first checked with a test script on about 35 sample questions, such as "CSE fees", "Block 8", "MH3" and "library timings", and the keyword search was adjusted where it picked the wrong page. The full bot, including the Gemini step, is tested on a phone with the questions below. Write Pass or Fail in the Result column after each test, then add the number of questions asked, the number answered correctly and the average response time.

TESTING:

No	Test Question	Expected behaviour
1	CSE fees	Tuition ₹1,95,000 and other fees ₹12,000 per year, with a "Based on" line
2	CSE subjects year 2	The year-2 subject list for CSE
3	Which is the cheapest course?	Mechanical, Civil, EEE and Aeronautical Engineering at ₹1,00,000 tuition
4	MH3	The MH3 hostel page
5	Which canteen has biryani?	Radha Krishnan Canteen (Andhra food)
6	Library timings	Says the timings are not added yet (Update soon); no invented hours
7	Where is Block 8?	Says the details are not added yet; no invented location
8	Hostel fees	Says hostel and mess fees are not added yet and to ask the Accounts Office
9	How do I prepare for GATE?	Study tips that start with "General advice (not official KARE information)"
10	Report an issue	Tells the student to use Report an Issue on the home screen (no AI call)
11	Any question with the internet off	Shows the stored text with a note that the AI could not answer

6.Conclusion & Future Scope

Conclusion: The KARE Help Bot connects the campus information stored in Cloud Firestore to a Gemini language model through retrieve, augment and generate steps. Its answers come from KARE's own records and name the pages they are based on. Missing information is reported as "Update soon" instead of being guessed, admins keep the knowledge current from inside the app, and a stored-text fallback keeps the bot useful when the AI service is unavailable. Together these give students one place to ask for campus information in plain words.

Future Scope:

- Vector embeddings and a vector database for semantic retrieval, so a question worded differently from the stored page still finds it.
- Ingestion of PDF documents such as the student handbook and regulations, split into chunks.
- Role-based answers, so students, staff and admins see the information meant for them.
- A thumbs up and thumbs down button on each answer to measure and improve retrieval.
- Questions in Tamil as well as English.